GS Quant Architect v4.02 - Variable Tuning Guide

This guide explains every input variable in `GS-QuantArchitectv6.mq5` so you can tune the EA with intention.

How to tune safely

• Optimize in stages (entry -> exits -> market filters).

• Keep fixed lot (`InpFixedLot=0.01`) while optimizing logic.

• Use real ticks and forward testing before increasing risk.

• Change only a few related variables per pass.

1) GLOBAL

• `InpMagicNumber` (default `20260306`)

- Unique ID for this EA's trades.

- Use a different value per strategy/account instance.

• `InpTradeComment` (default `"GSQ4"`)

- Comment tag for orders and diagnostics.

- Useful for filtering trades in history.

• `InpDebugMode` (default `true`)

- Enables detailed log output.

- Keep `true` while tuning, set `false` for clean production logs.

• `InpLogTrades` (default `true`)

- Logs open/close actions.

- Can be disabled during large optimization sweeps.

• `InpSignalTF` (default `PERIOD_M15`)

- Main decision timeframe.

- Lower TF = more trades/noise; higher TF = fewer trades/smoother behavior.

• `InpWarmupBars` (default `20`)

- Bars required before trading starts.

- Increase if indicators are unstable at start of test.

2) EXECUTION SAFETY

• `InpMaxSlippage` (default `30` points)

- Max acceptable price deviation on market orders.

- Tight values can miss entries; loose values can worsen fill quality.

• `InpSpreadFilterPoints` (default `0` = off)

- Blocks entries when spread exceeds threshold.

- Very important for forward robustness and news periods.

• `InpMinSecondsBetweenEntries` (default `30`)

- Cooldown between entries.

- Helps avoid rapid flip-flopping in chop.

• `InpAllowBuy` / `InpAllowSell` (default `true` / `true`)

- Direction gate switches.
- Useful for directional bias tests.

• `InpOnePositionPerDirection` (default `true`)

- If `true`, only one BUY and one SELL max at a time.
- Reduces over-stacking and duplicate exposure.

• `InpMaxOpenPositions` (default `2`)

- Absolute cap of simultaneous positions.
- Lower = safer; higher = more opportunity but more clustered risk.

3) RISK

• `InpRiskLevel` (default `RISK_MODERATE`)

- Base risk profile for dynamic lot sizing.
- Conservative/Moderate/Aggressive map to lower/higher risk percent.

• `InpFixedLot` (default `0.00`)

- `0` means risk-based position size.
- `>0` forces fixed lots (best for optimizer comparability).

• `InpMaxDailyDrawdown` (default `6.0%`)

- Daily circuit breaker for new entries.
- Tighten for capital preservation.

• `InpMaxTotalDrawdown` (default `20.0%`)

- Hard kill switch threshold.
- If reached, EA stops opening and can close all positions.

4) STRATEGY TOGGLES

• `InpEnableTrend` (default `true`)

- Includes trend-following score in entry decision.

• `InpEnableMeanReversion` (default `true`)

- Includes band/RSI reversion score.

• `InpEnableBreakout` (default `true`)

- Includes breakout structure score.

• `InpEnableMomentum` (default `true`)

- Includes RSI/MACD momentum score.

• `InpEnableVolatility` (default `true`)

- Includes ATR-regime score.

Tip: Turn off one module at a time to identify which signal family hurts PF.

5) INDICATORS

• `InpATRPeriod` (default `14`)

- Core ATR used for stop sizing and management.

• `InpEMAFast` / `InpEMASlow` / `InpEMATrend` (default `21/55/200`)

- Trend structure backbone.
- Larger gaps usually reduce noise but can delay entries.

• `InpRSIPeriod` (default `14`)

- RSI calculation period.
- Lower = faster, noisier.

• `InpRSIOverbought` / `InpRSIOversold` (default `68/32`)

- Reversion trigger sensitivity.
- More extreme values reduce trade count but can improve selectivity.

• `InpBBPeriod` / `InpBBDev` (default `20 / 2.0`)

- Bollinger context for mean reversion and volatility behavior.

• `InpADXPeriod` (default `14`)

- ADX smoothing period.

• `InpADXTrendMin` (default `18.0`)

- Minimum trend strength to trust trend continuation logic.
- Increasing often helps PF but lowers frequency.

• `InpBreakoutLookback` (default `24`)

- Window for breakout high/low detection.
- Larger values prefer stronger structural breaks.

• `InpATRFastPeriod` / `InpATRSlowPeriod` (default `7 / 50`)

- Volatility regime ratio.
- Bigger separation increases regime contrast.

6) ADAPTIVE ENTRY

• `InpEntryScoreBase` (default `35.0`)

- Main threshold to open a trade from composite score.
- Higher = stricter entries (usually higher PF, fewer trades).

• `InpEntryScoreMin` (default `18.0`)

- Minimum threshold floor after decay.
- Prevents threshold from becoming too loose.

• `InpThresholdDecayBars` (default `24`)

- Bars required before threshold decays.
- Smaller = threshold loosens faster after no-trade periods.

• `InpThresholdDecayStep` (default `5.0`)

- Amount threshold relaxes at each decay step.

- Too high can overtrade low-quality setups.

• `InpGuaranteeActivity` (default `true`)

- Allows forced entries when no trades for long periods.

- Improves activity but can hurt forward PF if too aggressive.

• `InpForceTradeBars` (default `24`)

- Bars without trades before forced entry logic can trigger.

• `InpForceRiskScale` (default `0.50`)

- Risk multiplier applied to forced trades.

- Lower is safer (`0.20` to `0.50` common for stability).

• `InpForceTradeNeedsTrend` (default `true`)

- Requires trend confirmation before forced entries.

- Strongly recommended for forward robustness.

7) OPTIMIZER TARGETS (used by OnTester Custom Max)

• `InpOptMinProfitFactor` (default `1.50`)

- Hard PF floor; runs below this are scored as zero in custom objective.

• `InpOptMinTrades` (default `80`)

- Minimum trade count floor for statistical relevance.

Note: These only affect `OnTester()` optimization score, not live entry logic.

8) EXIT / MANAGEMENT

• `InpSL_ATR_Mult` (default `1.8`)

- Stop-loss distance in ATR units.

- Higher = safer/wider stops, lower hit rate pressure, larger average loss.

• `InpTP_RR` (default `2.2`)

- Take-profit distance as multiple of SL distance.

- Higher values increase payoff asymmetry but reduce win rate.

• `InpTrailATR_Mult` (default `1.2`)

- ATR trailing distance.

- Tight trails lock profits quickly but can cut trends early.

• `InpBreakEvenATR` (default `1.0`)

- Profit threshold (in ATR) before moving SL toward BE.

• `InpMaxHoldBars` (default `280`, `0` disables)

- Time stop; closes stale positions.

- Useful to limit overnight regime drift exposure.

• `InpStrictProtectionMode` (default `true`)

- Requires proper SL/TP protection behavior.

- Recommended `true` for forward safety.

• `InpMaxUnprotectedSeconds` (default `15`)

- Emergency timeout for any position lacking SL/TP in strict mode.

9) SESSION FILTER

• `InpSessionFilter` (default `SESSION_ALL`)

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- Session gate:
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- `SESSION_ALL`, `SESSION_LONDON`, `SESSION_NEWYORK`, `SESSION_OVERLAP`, `SESSION_ASIAN`, `SESSION_NO_ASIAN`.

• `InpLondonStart` / `InpLondonEnd` (default `8/16`)

- London session boundaries.

• `InpNYStart` / `InpNYEnd` (default `13/21`)

- New York session boundaries.

• `InpAsianStart` / `InpAsianEnd` (default `0/8`)

- Asian session boundaries.

• `InpNoTradeFriday` (default `false`)

- Blocks late-Friday entries to reduce weekend-gap vulnerability.

Practical tuning playbook

1. **Base edge pass**

- Optimize: `InpEntryScoreBase`, `InpEntryScoreMin`, `InpThresholdDecayBars`, `InpTP_RR`.

2. **Exit quality pass**

- Optimize: `InpSL_ATR_Mult`, `InpTrailATR_Mult`, `InpBreakEvenATR`, `InpMaxHoldBars`.

3. **Market robustness pass**

- Optimize: `InpSessionFilter`, `InpSpreadFilterPoints`, `InpADXTrendMin`.

4. **Forward hardening**

- Keep `InpStrictProtectionMode=true`, prefer `InpGuaranteeActivity=false` if forward cliff appears.

5. **Scaling**

- Increase lot size only after stable forward PF/DD.

- Lot size changes returns and DD, not the core PF ratio.

Suggested safe profiles

• **PF-focused**: higher `InpEntryScoreBase`, `InpGuaranteeActivity=false`, tighter session/spread filters.

• **Frequency-focused**: lower entry thresholds, but keep strict protection and moderate spread filter.

• **Forward-stability**: `MaxOpenPositions=1`, `SESSION_NO_ASIAN`, `InpNoTradeFriday=true`, stricter DD caps.